

Operating Systems

BRANCH: COMPUTER SCIENCE (CSE)

COURSE CODE: CS302

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Operating Systems (CS302) under the Computer Science (CSE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Process Management

Process concept, process states, PCB, Process scheduling (FCFS, SJF, SRTF, Round Robin, Priority). Inter-process communication (IPC) and synchronization. Semaphores, Monitors, and Classic IPC problems.

Unit 2: Memory Management

Logical vs Physical address spaces, Swapping, Contiguous memory allocation, Paging, Segmentation. Virtual Memory: Demand Paging, Page replacement algorithms (FIFO, LRU, Optimal), Thrashing.

Unit 3: Storage & File Systems

File concept, Access methods, Directory structure, File system mounting, File sharing, and Protection. Allocation methods (Contiguous, Linked, Indexed), Free-space management. Disk scheduling (FIFO, SSTF, SCAN, C-SCAN).

Unit 4: Deadlocks & Security

System model, Deadlock characterization, Deadlock prevention, Deadlock avoidance (Banker's Algorithm), Deadlock detection and recovery. Protection & Security principles.

SJF is optimal for minimum average waiting time. Page Fault Rate, Effective Access Time (EAT) equations.

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



